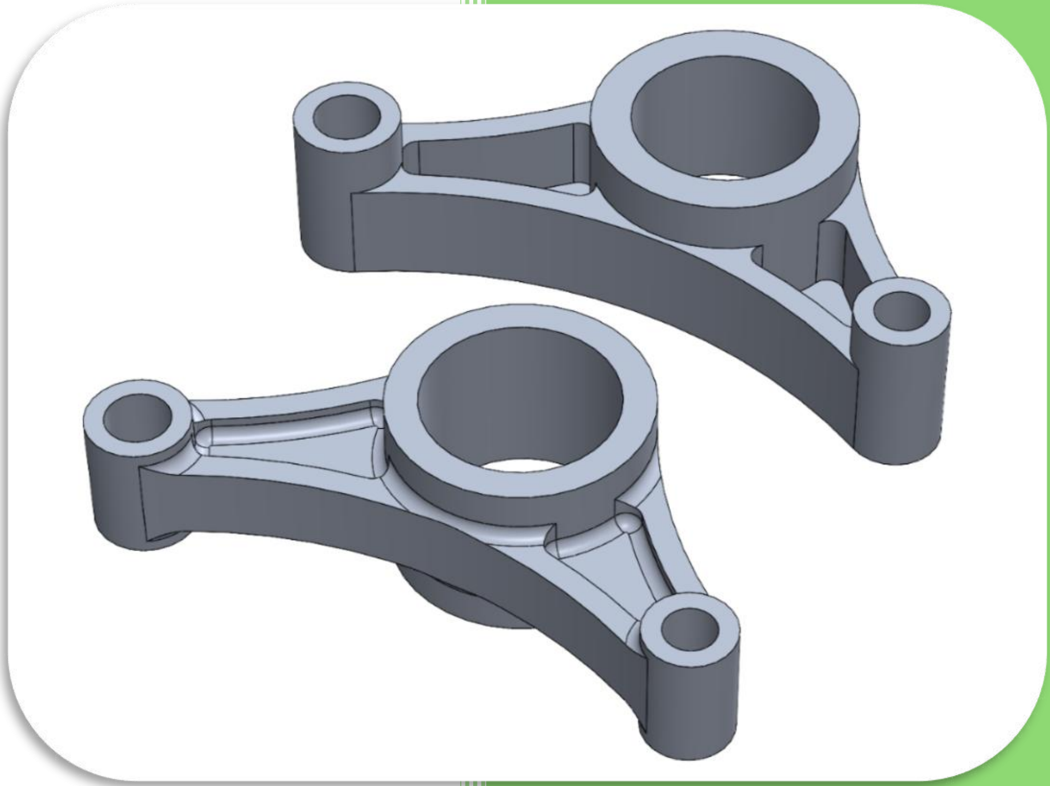
MMAE 445

# Bracket Design and Manufacturing Study



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# Introduction

This case study goes through the complete computer aided design (CAD) and Mastercam workflow used to model, evaluate, document, and program two similar brackets, identified as Bracket P-1 and Bracket P-2. The study starts with the modeling of each of the brackets in SolidWorks before conducting finite element analysis (FEA) studies to determine points of failure and factor of safety. A fixture is then created using a vice and riser assembly to support the manufacturing of the part. This fixture is then used in Mastercam to program the first and second operations for both Bracket P-1 and Bracket P-2. Once each operation is programmed, it is verified and compared with the expected model to ensure accuracy and highlight any shortcomings in the manufacturing process. Detailed screenshots are provided to give a clear representation of the workflow used in the case study.

# Bracket P-1 CAD & FEA

Prior to modeling Bracket P-1 and Bracket P-2, the global document units were set to inches, pounds, and seconds (IPS) and the dimension precision was set to three decimal places (0.XXX).

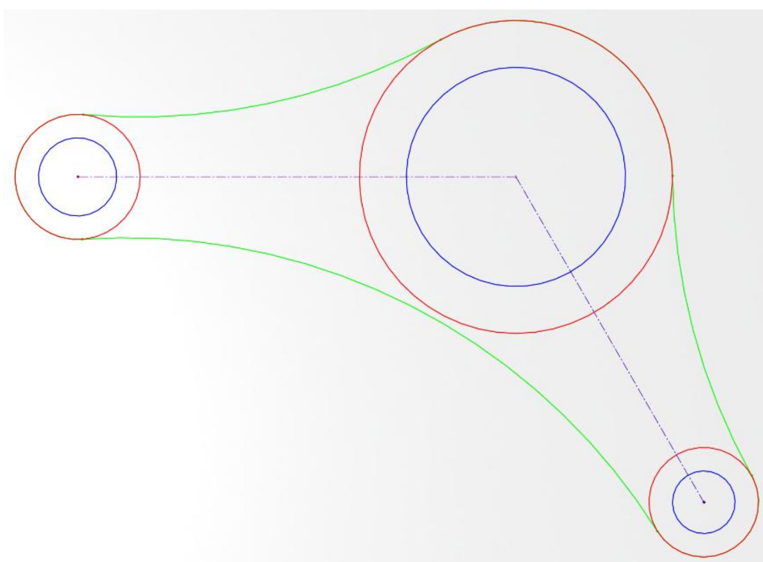


***Figure 1: Bracket P-1 parent sketches***

Three sketches were made to allow for the independent adjustment of the key part features. They were set up as parent sketches to allow for the quick adjustment of one sketch that propagates through the rest of the part. The lines in blue (S1) are the inner circles, the lines in red (S2) are the outer circles, and the lines in green (S3) are the outside profile. The outer circles and outside profile overlap in Figure 1, meaning within SolidWorks the outside profile is a closed sketch.

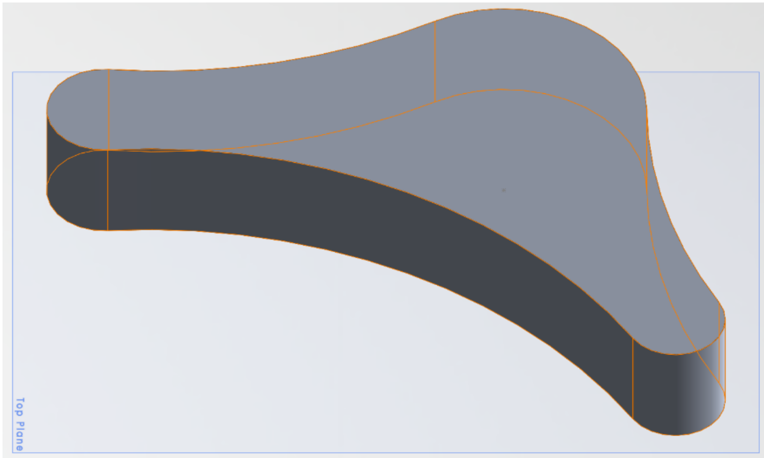


***Figure 2: Boss extrude using outside profile sketch***

The second step was to extrude boss/base the outside profile as seen in Figure 2. This was done by creating a new sketch and using convert entities on S3 to create a parent child relationship. Now, whenever a change is made to S3, it is automatically changed in this new sketch.

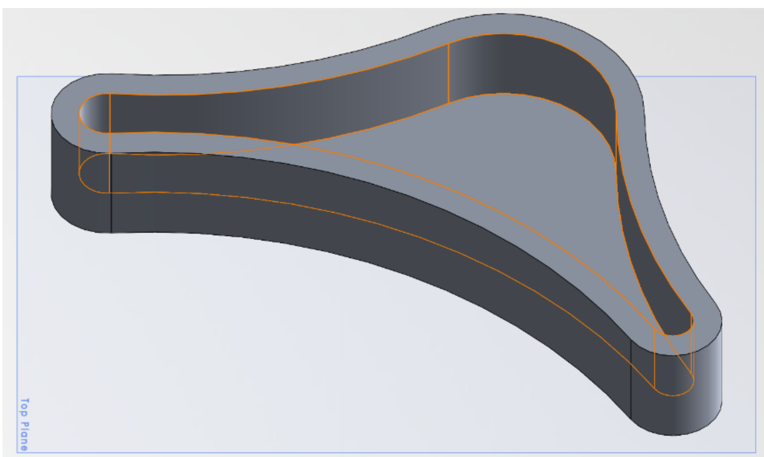


***Figure 3: Shell command***

The third step was to use the shell command to hollow out the inside of the extrusion, as seen in Figure 3. This was done to create the walls and floor of the final bracket.

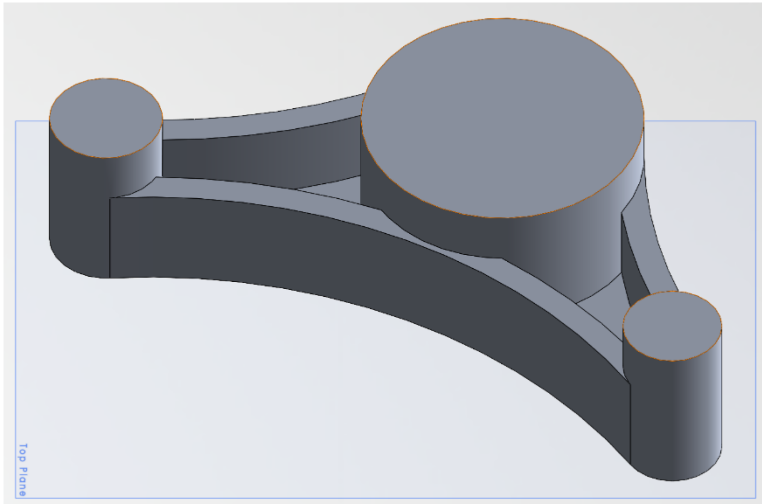


***Figure 4: All circles extrusion***

The fourth step was to create a new sketch and extrude the circles found in S2. A similar sketch process to step two was done to create a parent child reference.

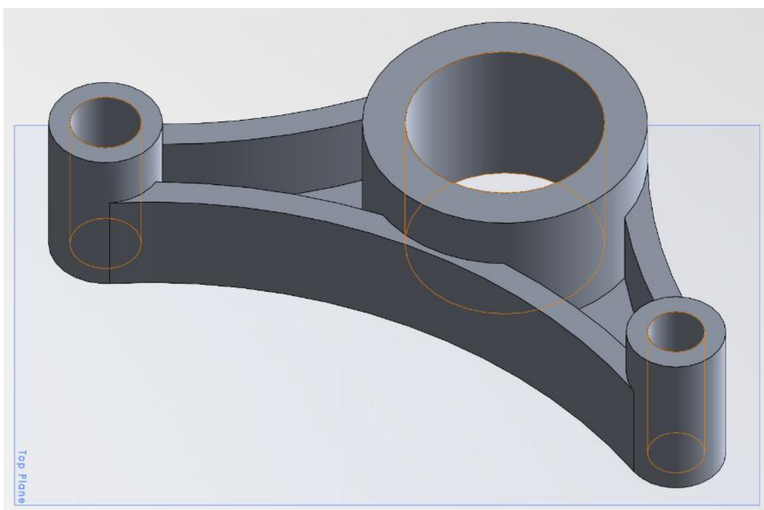
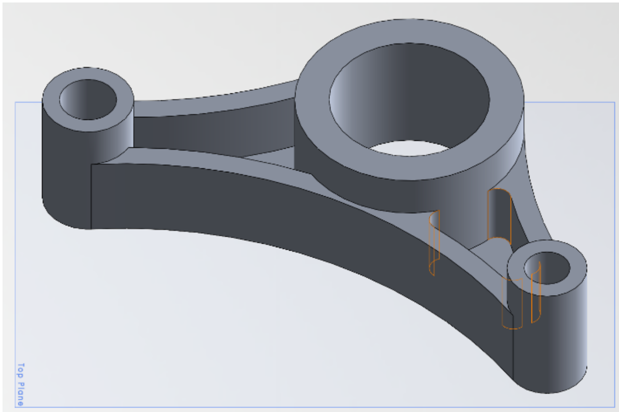


***Figure 5: Circle center extrude cut***

The fifth step was to use S1 to perform an extrude cut to clear out the brackets mounting holes as seen in Figure 5. In the setup of the extrude cut “through all” was used so that regardless of any extrusion height changes to the mounting points, the hole always goes through.



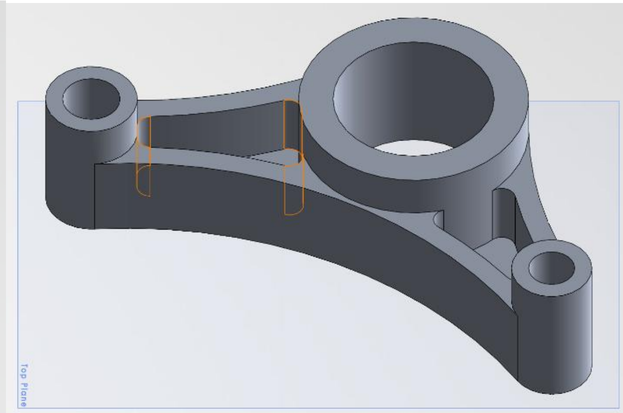


*Figure 6: Pocket fillet* *Figure 7: Pocket fillet*

The final step was to add fillets to the inside corners of both pockets as seen in Figures 6 and 7. This was done to reduce the stress concentration caused by sharp corners.

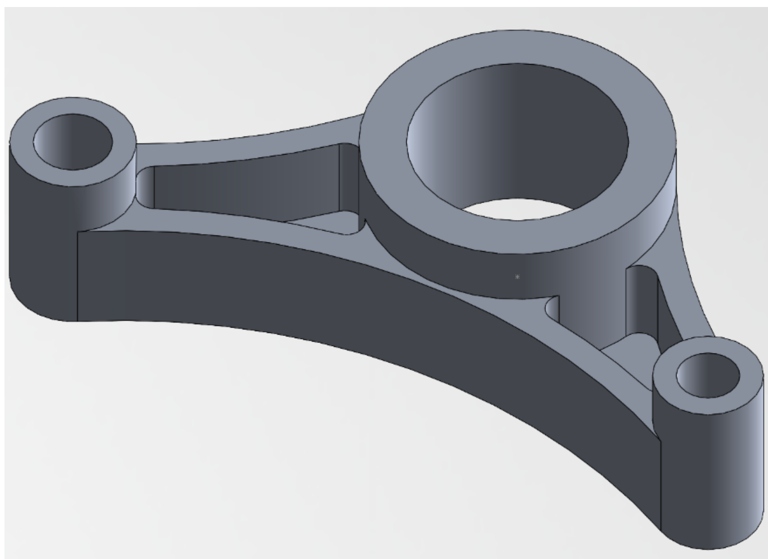


*Figure 8: Bracket P-1 final model*

Once the full model had been made, material was applied so that mass properties could be calculated and FEA could be done. The material chosen was 6061-T6 Aluminum, giving the following properties:

| Mass (Pounds) | Volume (Cubic Inches) | Surface Area (Square Inches) |
|---:|---:|---:|
| 0.0955 | 0.9790 | 14.7472 |

*Table 1: Bracket P-1 properties*

The FEA for Bracket P-1 was performed in SolidWorks Simulation to evaluate its structural response under a 500 pound directionally applied load.

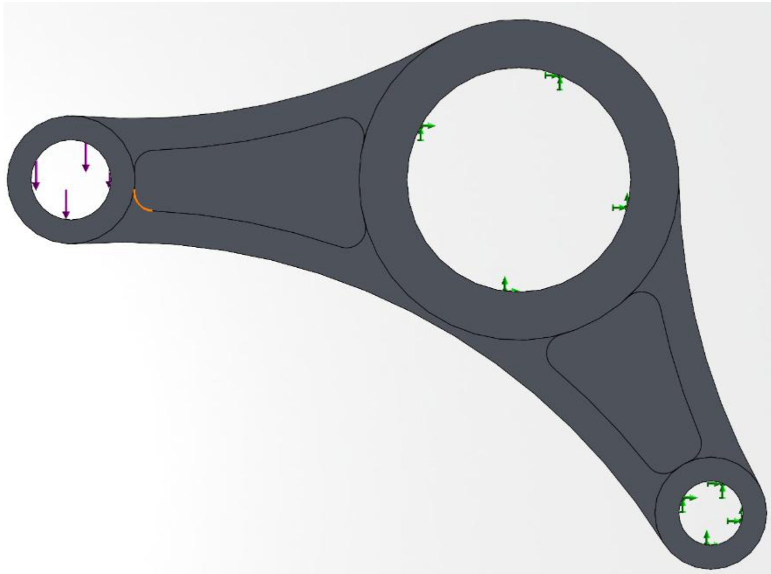


*Figure 9: Fixtures and loading direction for Bracket P-1 FEA*

As seen in Figure 9, the model was fixed in place at the right two holes, and the 500-pound load was applied to the left hole, parallel to the top plane. Once the fixtures and load were set, standard meshing was done and the study was run, giving results for stress, displacement, and factor of safety.

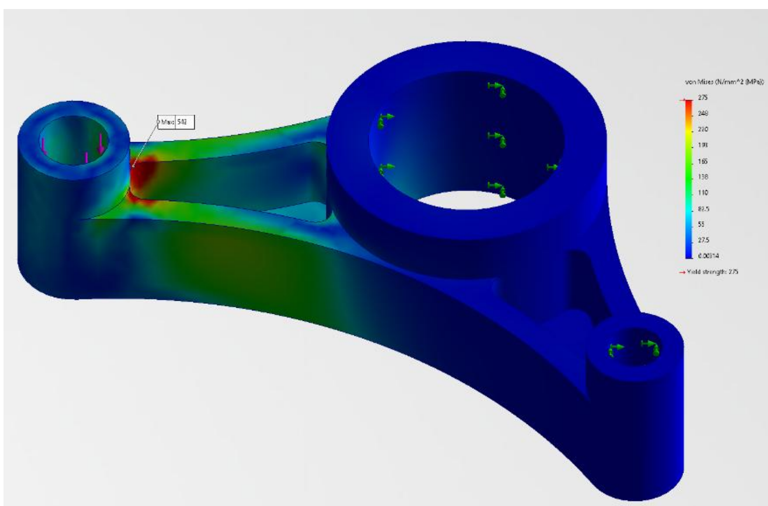



*Figure 10: Bracket P-1 stress analysis*

From Figure 10, the maximum stress, shown on the left, was 542 MPa. This is significantly higher than 6061-T6 Aluminums yield strength of 275 MPa, indicating part failure.

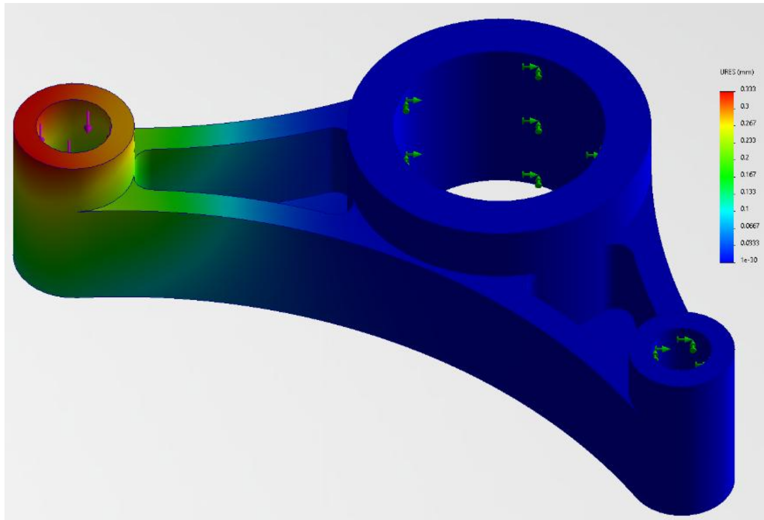


***Figure 11: Bracket P-1 displacement analysis***

Figure 11 displays the region of maximum flexural displacement due to the loading condition. As expected, the left side of the bracket, which was not fixed in place, experienced the most displacement, up to a maximum deflection of 0.0131 inches on the left outside edge of the left circle.

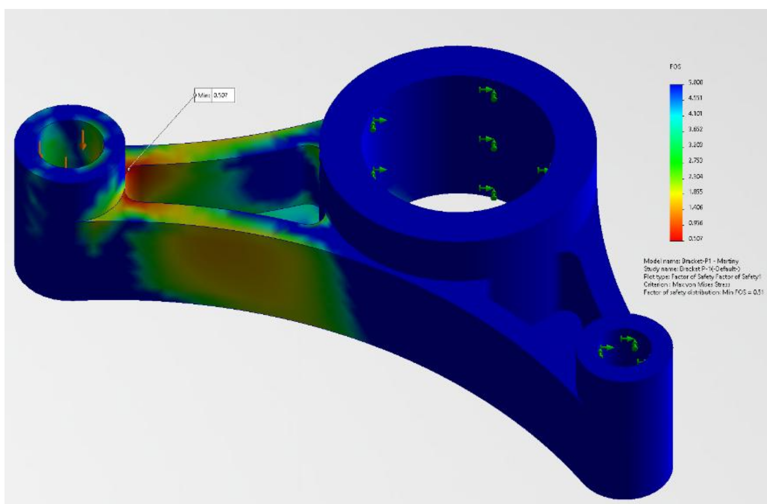


***Figure 12: Bracket P-1 factor of safety analysis***

To determine if a part is safe to use, engineers will build in or calculate the factor of safety (FOS). In this case, the FOS is based on the material's yield strength and working strength, shown in the following formula.

$$\frac{\sigma_{yield}}{\sigma_{loaded}} = FOS \tag{1}$$

Using $\sigma_{yield} = 275\ MPa$ and $\sigma_{loaded} = 542\ MPa$, the calculated and displayed FOS is 0.507. This means that the part will fail and is not safe to use in its current configuration. Figure 12 shows where this minimum FOS occurs, which is exactly the same area as the maximum stress in Figure 10. A maximum FOS of 5 was chosen arbitrarily to highlight the general region of failure in the part, which is the left arm of the bracket, where it connects to the left mounting circle.

## Bracket P-2 CAD & FEA

To start Bracket P-2, Bracket P-1 was copied and edited, largely staying the same except for a few modifications to the parent sketches, and a modified modeling workflow.

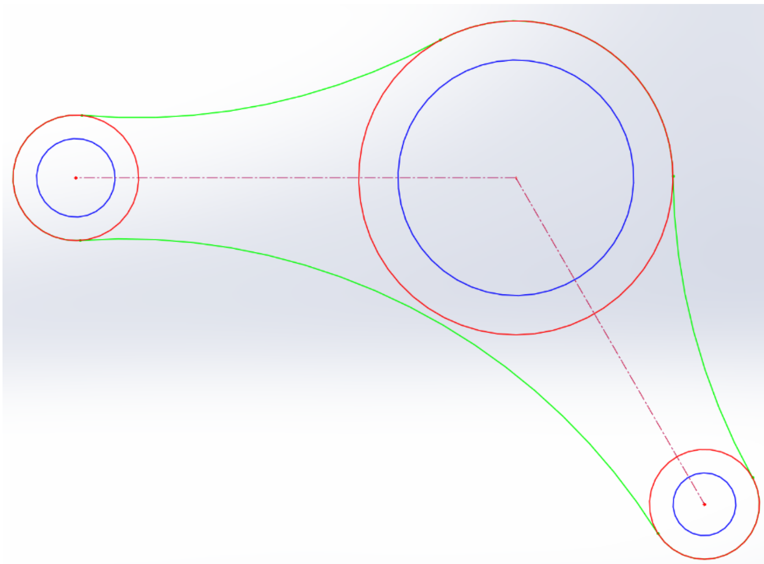


*Figure 13: Bracket P-2 parent sketches*

The three sketches from Bracket P-1 were edited to match the new dimensions for Bracket P-2, primarily the large hole in the center became slightly bigger. The lines in blue (S1) are the inner circles, the lines in red (S2) are the outer circles, and the lines in green (S3) are the outside profile. The outer circles and outside profile overlap in Figure 13, meaning within SolidWorks the outside profile is a closed sketch. The outside profile of Bracket P-1 and Bracket P-2 are the same, which will be useful for fixture making and manufacturing later. To speed up the creation of Bracket P-2, the rollback feature in the SolidWorks design tree was used to go back and edit each modeling step without having to completely redo everything manually.

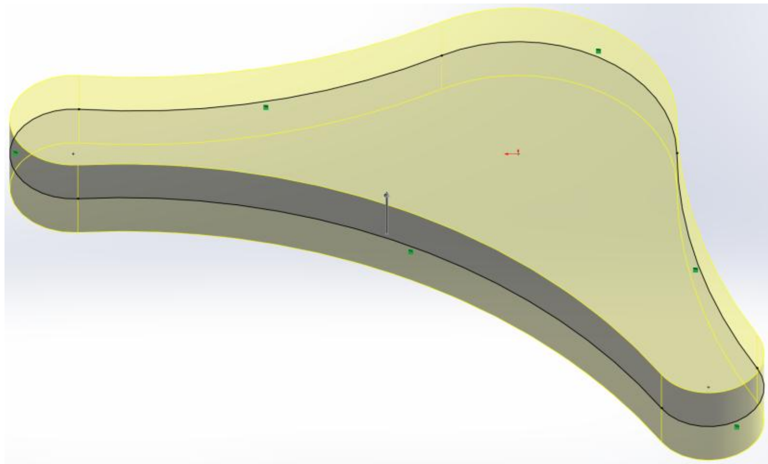


***Figure 14: Bracket P-2 midplane boss extrude***

The second step was to extrude boss/base the outside profile as seen in Figure 14. This was done by changing the boss/base extrusion from a blind extrude to a midplane extrude. Bracket P-2 is symmetrical across its midplane, so a midplane extrude allows for easy symmetry.

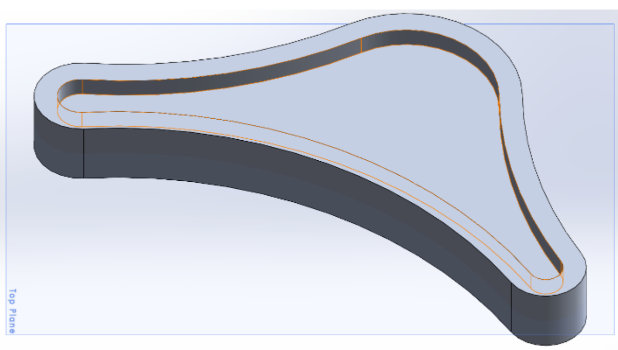


***Figure 15: Top surface shell command***

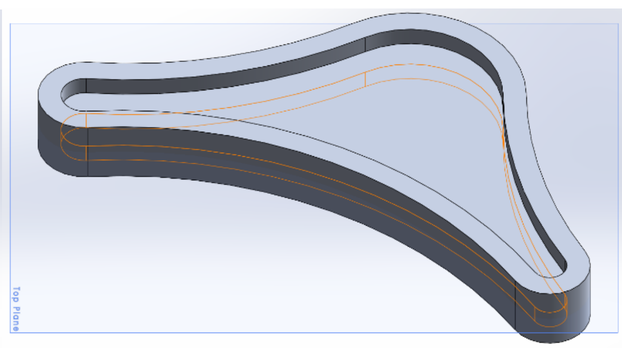


***Figure 16: Bottom surface shell command***

The third step was to shell out the top and bottom of the extrusion to set the wall and floor thickness as seen in Figures 15 and 16. The top and bottom needed to be done independently so that the floor was set at the midplane.

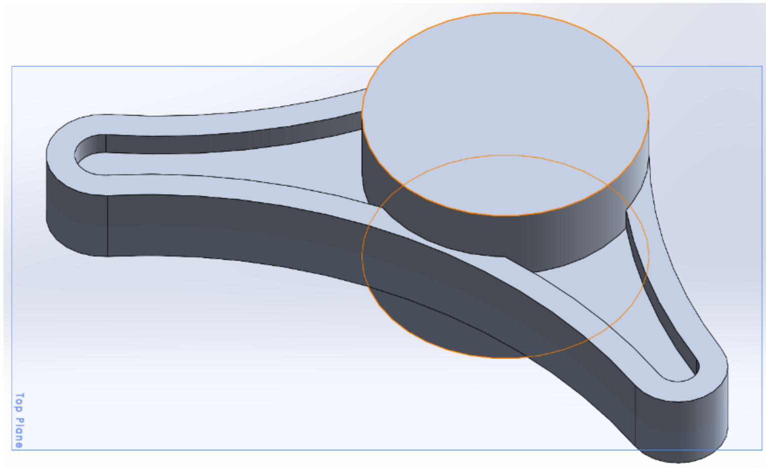


*Figure 17: Large hole extrusion*

The fourth step was to extrude the large hole as seen in Figure 17. This was done separately from the outside holes as they no longer share the same height. This was also done as a midplane extrude.

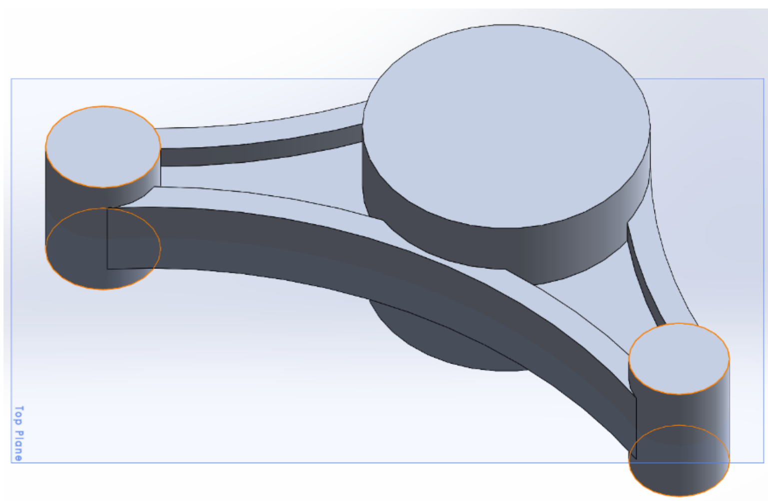


*Figure 18: Small outside hole extrusions*

The fifth step was to extrude the small outside holes as seen in Figure 18. They share the same total height and could be done together as a midplane extrusion.

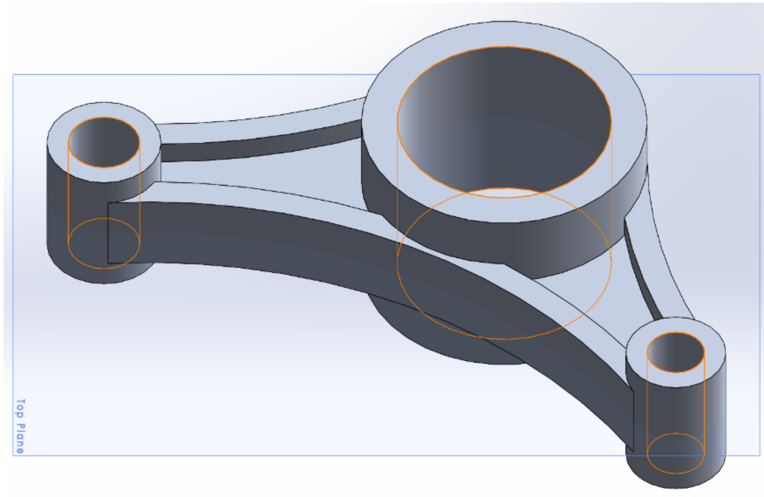



***Figure 19: Circle center extrude cut***

The sixth step was to use S1 to perform an extrude cut to clear out the brackets mounting holes as seen in Figure 19. A bi-directional through all extrude was used because the sketches are located at the midplane of the part.

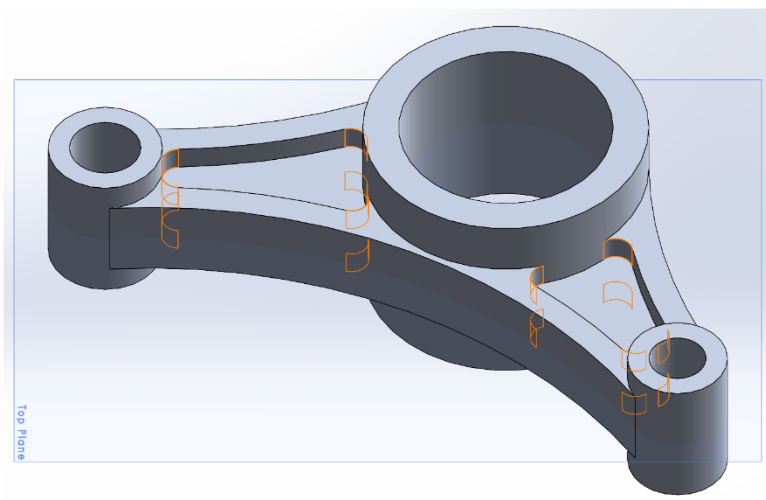



***Figure 20: Corner radius fillets***

The seventh step was to apply the corner fillets to all the pocket corners to limit stress concentration. The top and bottom are identical as seen in Figure 20.

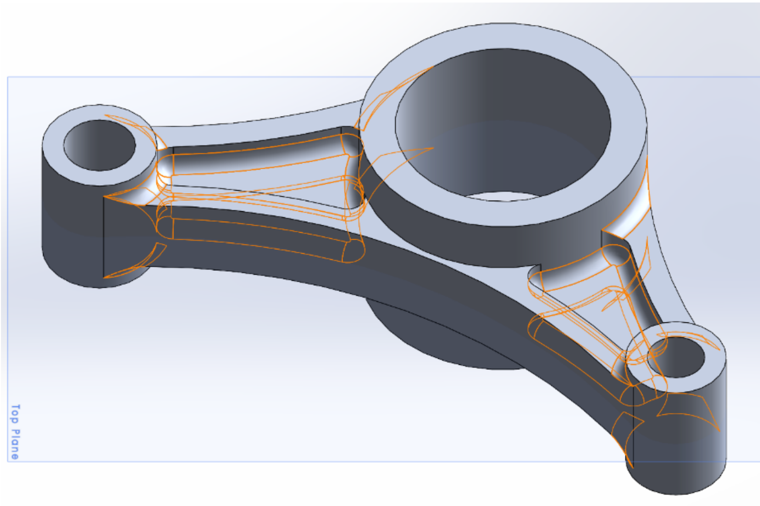


*Figure 21: Floor radius fillets*

The eighth step was to apply floor fillets to the bases of the outside mounting holes, as well as the inside edges of all the pockets, as seen in Figure 21.

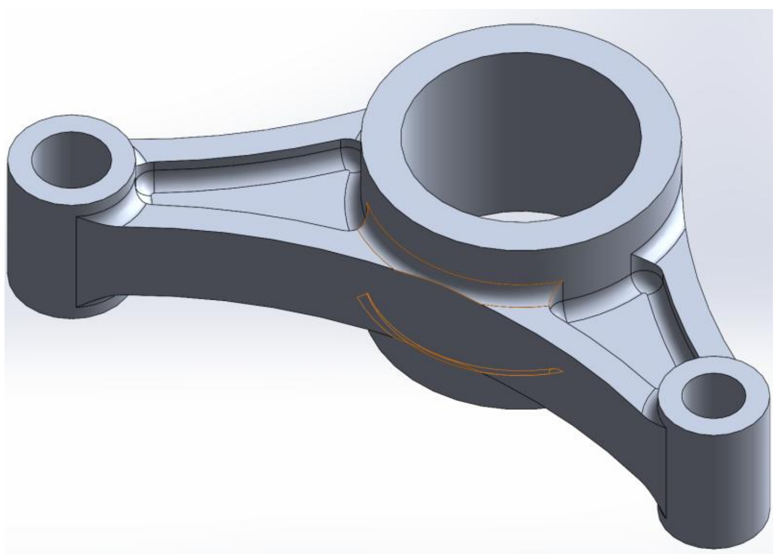


*Figure 22: Big circle floor fillet*

The ninth step was to apply the floor fillet to the remaining edge of the center mounting hole, as seen in Figure 22. It needed to be done separately because of the way SolidWorks generates the fillets normally. It can be seen that the bottom part of the fillet does not line up exactly with the floor. When using a standard fillet, this would cause the top edge of the fillet to not be continuous, but by doing it in a separate step “keep surface” can be used to create a continuous fillet.

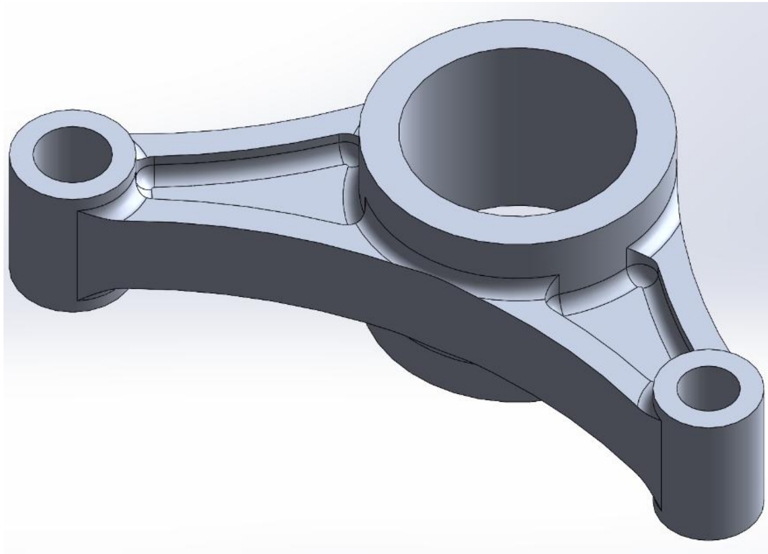


*Figure 23: Bracket P-2 final model*

Once the full model had been made 6061-T6 Aluminum was applied and the material properties were calculated and FEA could be done. The following are the calculated properties.

| Mass (Pounds) | Volume (Cubic Inches) | Surface Area (Square Inches) |
|---|---|---|
| 0.0888 | 0.9104 | 13.7631 |

*Table 2: Bracket P-2 properties*

The FEA for Bracket P-2 was performed in SolidWorks Simulation to evaluate its structural response under a 500 pound directionally applied load.

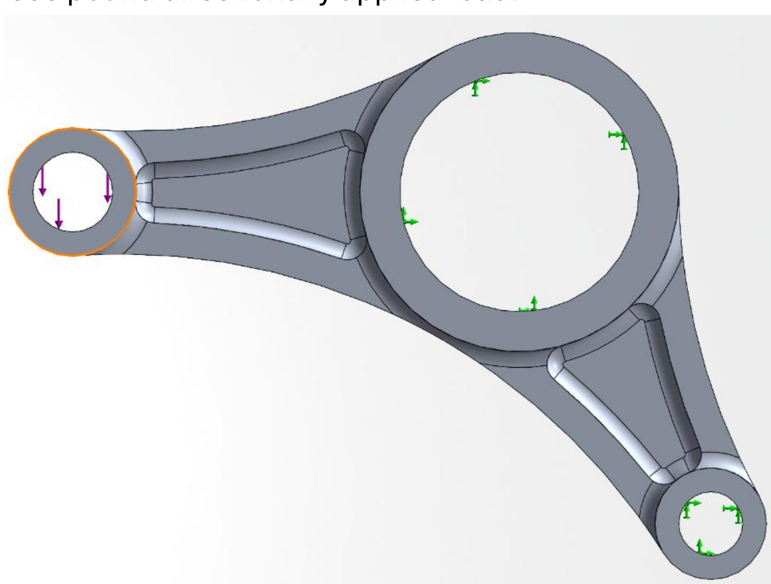


*Figure 24: Fixtures and loading direction for Bracket P-2 FEA*

Similar to Bracket P-1, the model was fixed in place at the right two holes, and the 500-pound load was applied to the left hole, parallel to the top plane, as seen in Figure 24. A standard mesh was created and the study was run.

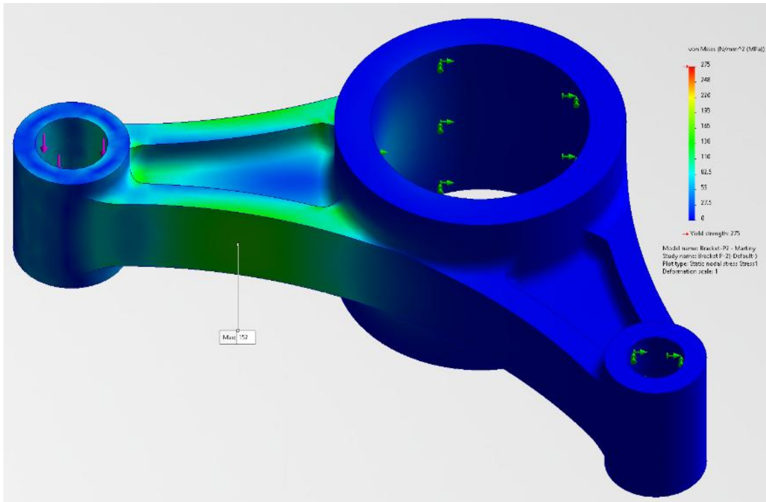



*Figure 25: Bracket P-2 stress analysis*

From Figure 25, the maximum stress, shown in the middle left, was 152 MPa. This is lower than the yield strength of 6061-T6 Aluminum, indicating the part is sufficiently strong under this loading condition.

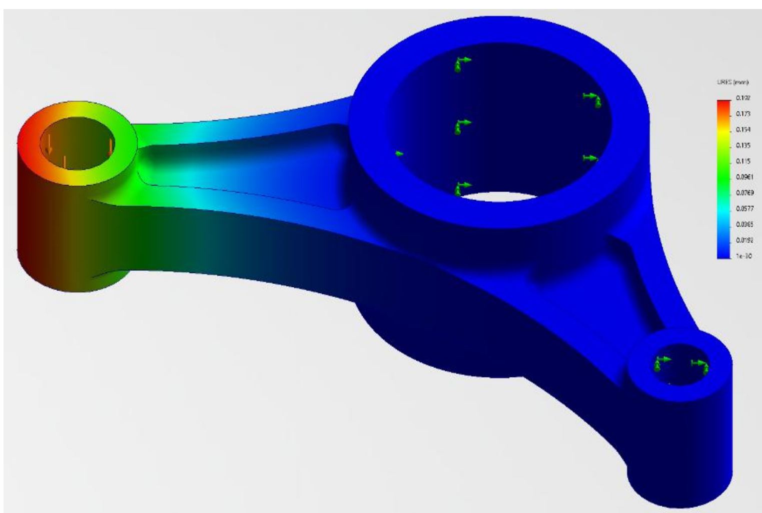



*Figure 26: Bracket P-2 displacement analysis*

Figure 26 displays the region of maximum flexural displacement due to the loading condition. As expected, the left side of the bracket, which was not fixed in place, experienced the most displacement, up to a maximum deflection of 0.00757 inches on the left outside edge of the left circle.

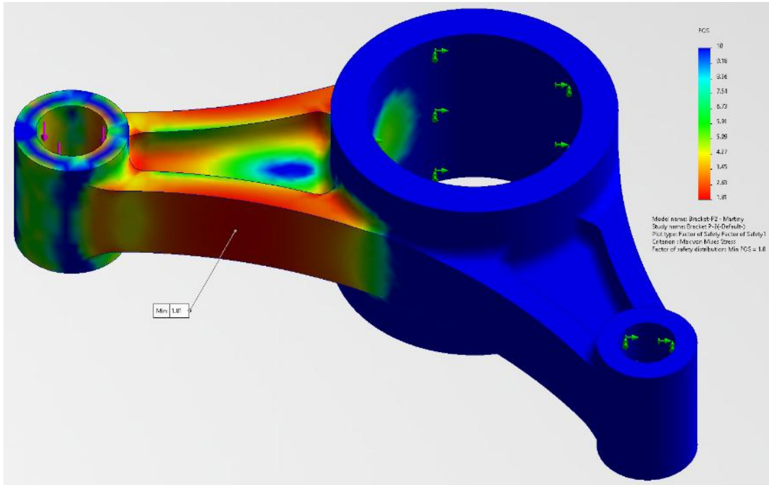


*Figure 27: Bracket P-2 factor of safety analysis*

Using Equation 1, the FOS was calculated with $\sigma_{yield} = 275\ MPa$ and $\sigma_{loaded} = 152\ MPa$. This gave a FOS of 1.815. This means the part is safe up to 1.815 times the current loading. To find the maximum load the part can take, the following relationship can be used.

$$\frac{\sigma_{loaded}}{F_{actual}} = \frac{\sigma_{yield}}{F_{max}} \tag{2}$$

With:

$$\sigma_{loaded} = 152\ MPa$$
$$\sigma_{yield} = 275\ MPa$$
$$F_{actual} = 500\ pounds$$
$$F_{max} = Maximum\ load\ before\ failure$$

Using Equation 2, $F_{max} = 904.61$ pounds. This can also be calculated using the FOS by multiplying the applied load by 1.81. A maximum FOS of 10 was chosen arbitrarily to highlight the region most likely to fail, which is the left arm of the bracket.

## Bracket P-1 & P-2 Comparison

| | Mass (Pounds) | Maximum Stress (MPa) | FOS |
|---|---|---|---|
| Bracket P-1 | 0.0955 | 542 | 0.507 |
| Bracket P-2 | 0.0888 | 152 | 1.815 |

*Table 3: Bracket P-1 & P-2 mass and FEA results*

Bracket P-2 shows a clear improvement over Bracket P-1 in terms of mass, maximum stress, and FOS, as seen in Table 3. Although Bracket P-2 is lighter at 0.0888 pounds compared to 0.0955 pounds for Bracket P-1, it performs significantly better under the same 500-pound loading condition. Bracket P-1 experiences a maximum stress of 542 MPa which is far above the yield strength of 6061-T6 Aluminum at 275 MPa, meaning the part fails and only achieves a FOS of 0.507. By comparison, Bracket P-2 experiences a maximum stress of 152 MPa, well below the aluminum yield strength, giving a FOS of 1.815. This FOS allows Bracket P-2 to be loaded to 904.61 pounds safely. The improvements from Bracket P-1 to Bracket P-2 are primarily due to the shift from unidirectional modeling to midplane modeling, as well as the inclusion of both corner and floor fillets, which reduce corner stress concentrations and help distribute the load throughout the part more consistently.

## Bracket P-1 & P-2 Drawings & GD&T

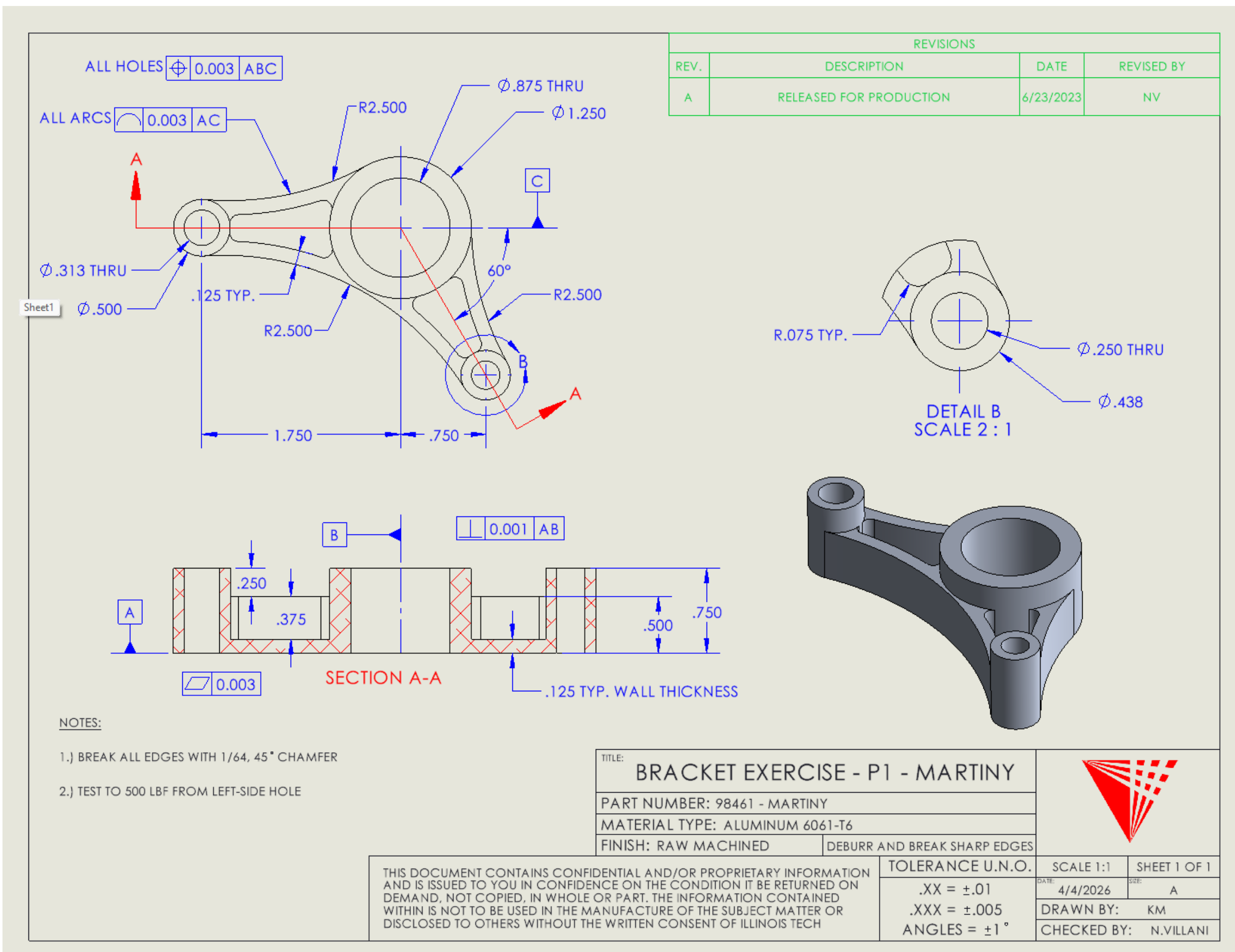



*Figure 28: Bracket P-1 reference drawing*

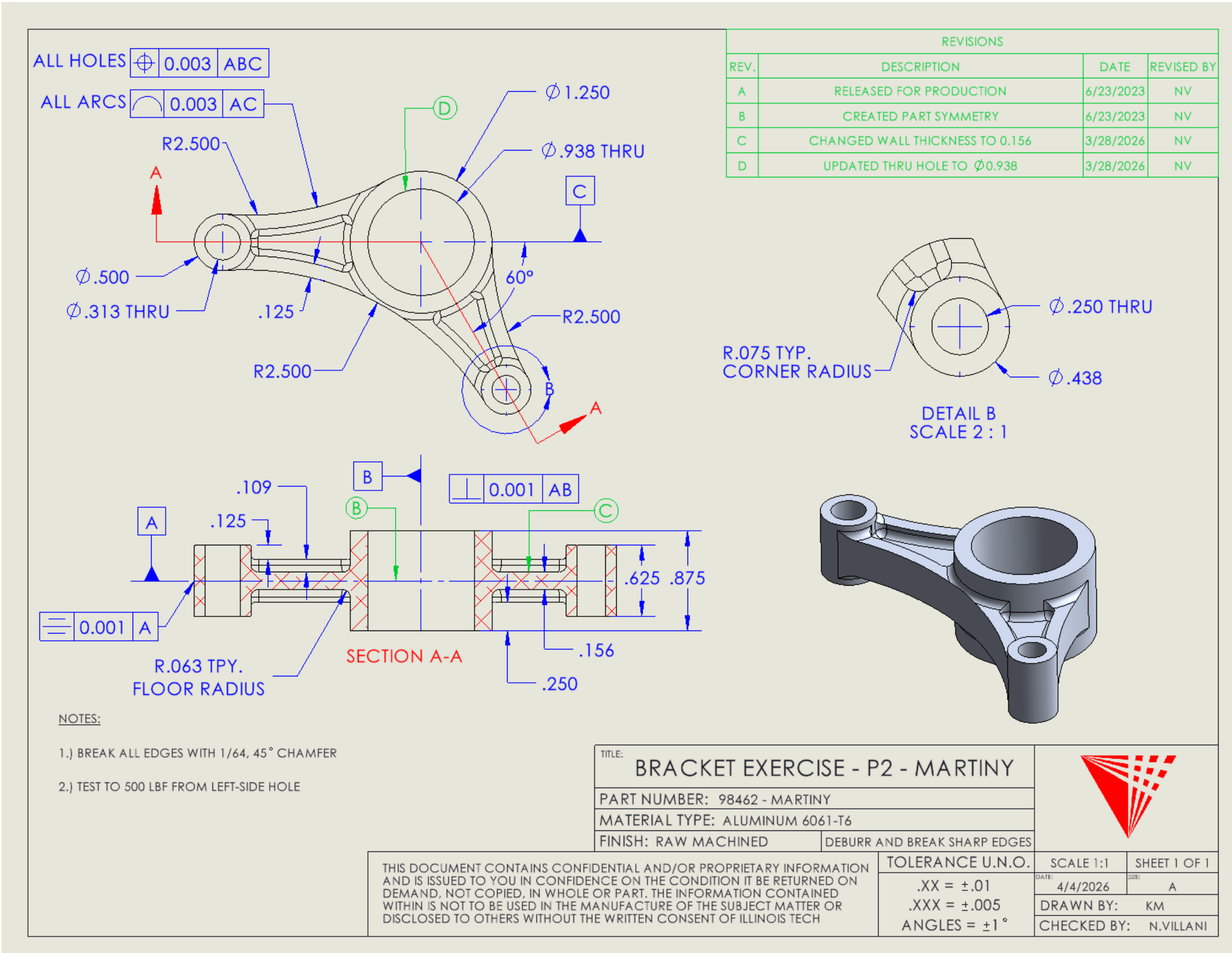



***Figure 29: Bracket P-2 reference drawing***

There are four views that are used for the drawings in Figures 28 and 29. They are a front orthographic view, a section view, a detail view, and a shaded isometric view. The front orthographic view gives the overall shape of the part and gives a clear picture of the major geometric details like the mounting holes and wall thickness. Section A-A is used to give an internal picture of both parts and to allow for the dimensioning of internal features like floor thickness, stepped geometry, and internal radii. These features would not be clearly visible in other exterior and are necessary in this case because both brackets are somewhat hollow and have recessed features. Detail B is used to enlarge small details like the corner radius and small hole dimensions. These dimensions could be added to other views but would result in crowding. The shaded isometric view gives an overall impression of how the part is supposed to look when it's done. It is not used for any dimensioning or tolerancing but is useful for manufacturing.

| Symbol | Name | Type | Description | Active Drawing |
|---|---|---|---|---|
| ⌖ | Position | Location | *A position details the location of a hole, slot, or circular feature* | P-1, P-2 |
| ⌒ | Profile of a Line | Profile | The shape of a 2D cross section | P-1, P-2 |
| ⊥ | Perpendicularity | Orientation | A 90-degree orientation between features | P-1, P-2 |
| ⏥ | Flatness | Form | Flatness states that an entire surface lies in one plane | P1 |
| ⌯ | Symmetry | Location | Symmetry establishes a center plane and makes points on each side of the plane equal | P2 |

***Table 4: GD&T symbols used in Figures 28 and 29***

Each of the geometric dimensioning and tolerancing (GD&T) symbols used in both Bracket P-1 and P-2 drawings play an important role in determining how the part will function after it has been manufactured. Position is used to declare the exact position of the holes because their location determines how the final part works with other parts. The holes might have the correct diameter, but if their centers are not correctly placed, there could be assembly issues after manufacturing. Profile of a line is used on the arcs because it is important for the inner and outer curves to maintain their shape. Perpendicularity is used to keep a feature at a square orientation relative to the chosen datums. Flatness is used to keep a surface planar and makes sure that the contact between the part and another surface or part is even, with very little warping. Symmetry is used specifically in P-2 because the part was designed around the midplane, meaning that across the midplane, every point needs to be identical. Having symmetry ensures the part is balanced and loads are distributed evenly.

The GD&T tolerances are different from the tolerances called out in the title block because they control specific geometric requirements that a general size tolerance cannot. The title block tolerance gives a default, general value for the allowed variation in dimensions, but does not control the specifics of a feature that might be required for proper function. GD&T is used on performance critical features where the function, fit, and manufacturability matter more than the general sizing alone. For Bracket P-1 and P-2 drawings, the tighter tolerances make sure that the holes align correctly, the arcs maintain their intended shape, and important surface features remain properly oriented relative to each other and defined datums.

# Vice & Riser Assembly & Drawing

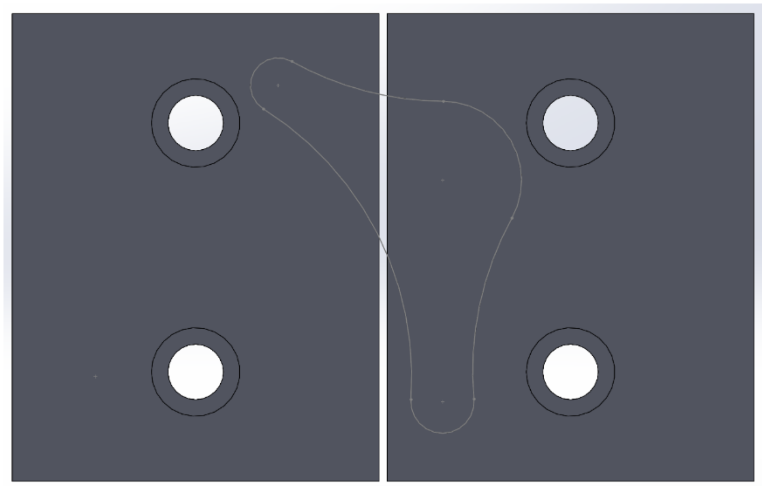


***Figure 30: Soft jaws prior to parent-child relationship with bracket profile***

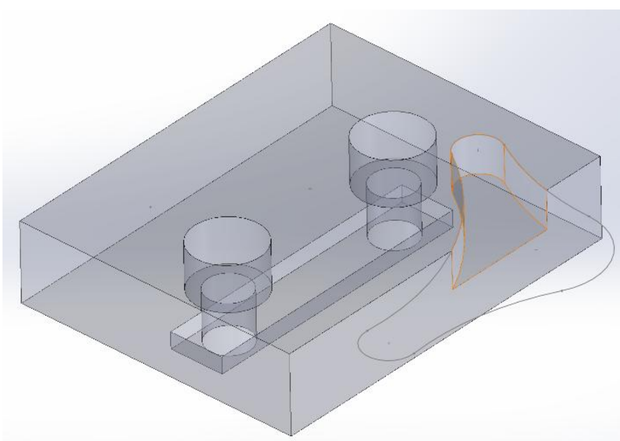


***Figure 31: Front soft jaw with bracket profile cut out***

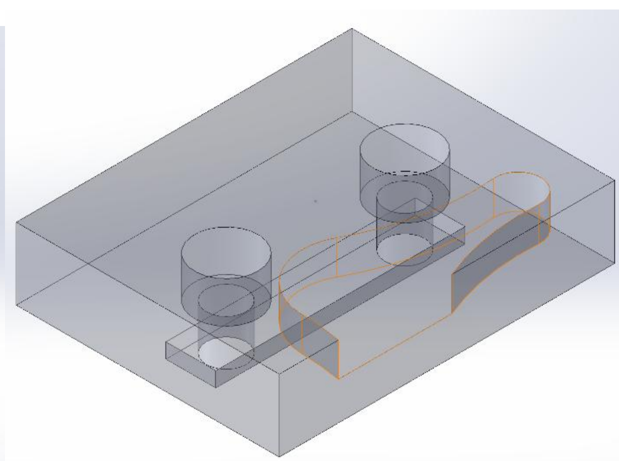


***Figure 32: Back soft jaw with bracket profile cut out***

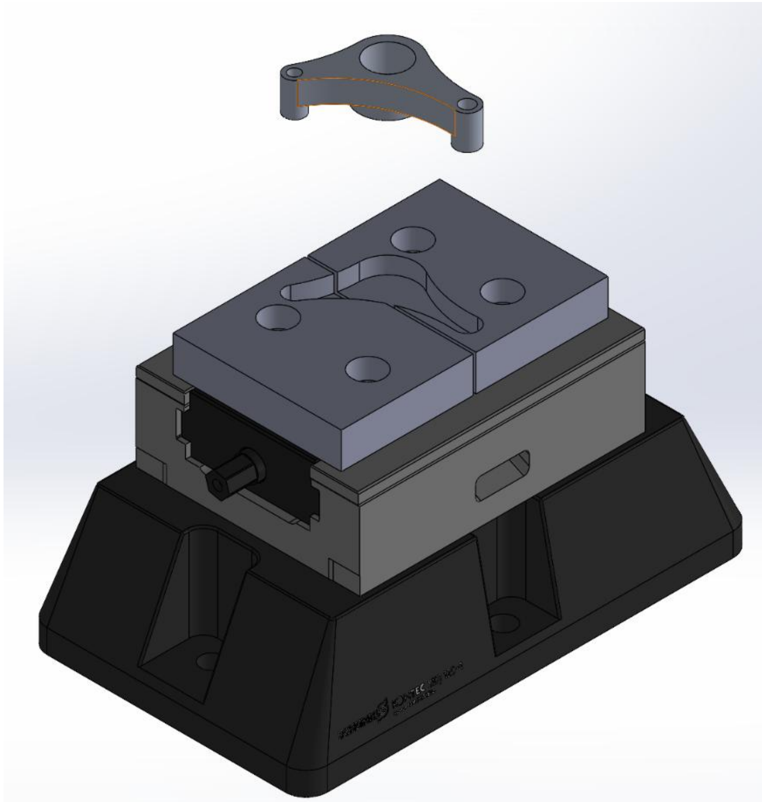


*Figure 33: Vice and riser with parent-child relationship on the soft jaws*

A parent-child relationship is when one feature depends on another feature, sketch, or component for positioning, shape, or definition. If the parent file or sketch changes, that change is then propagated through each of the child sketches, features, or components automatically. In the vice and riser assembly, there is a parent-child relationship between the bracket and soft jaws. On the soft jaws, the profile of the bracket is used to make a pocket that can be used in the manufacturing of Bracket P-1 and P-2. This is done through linked sketches, where the bracket serves as the parent, and both the jaws are the children. As a result, if the bracket profile geometry changes, this change is automatically updated on the soft jaws.

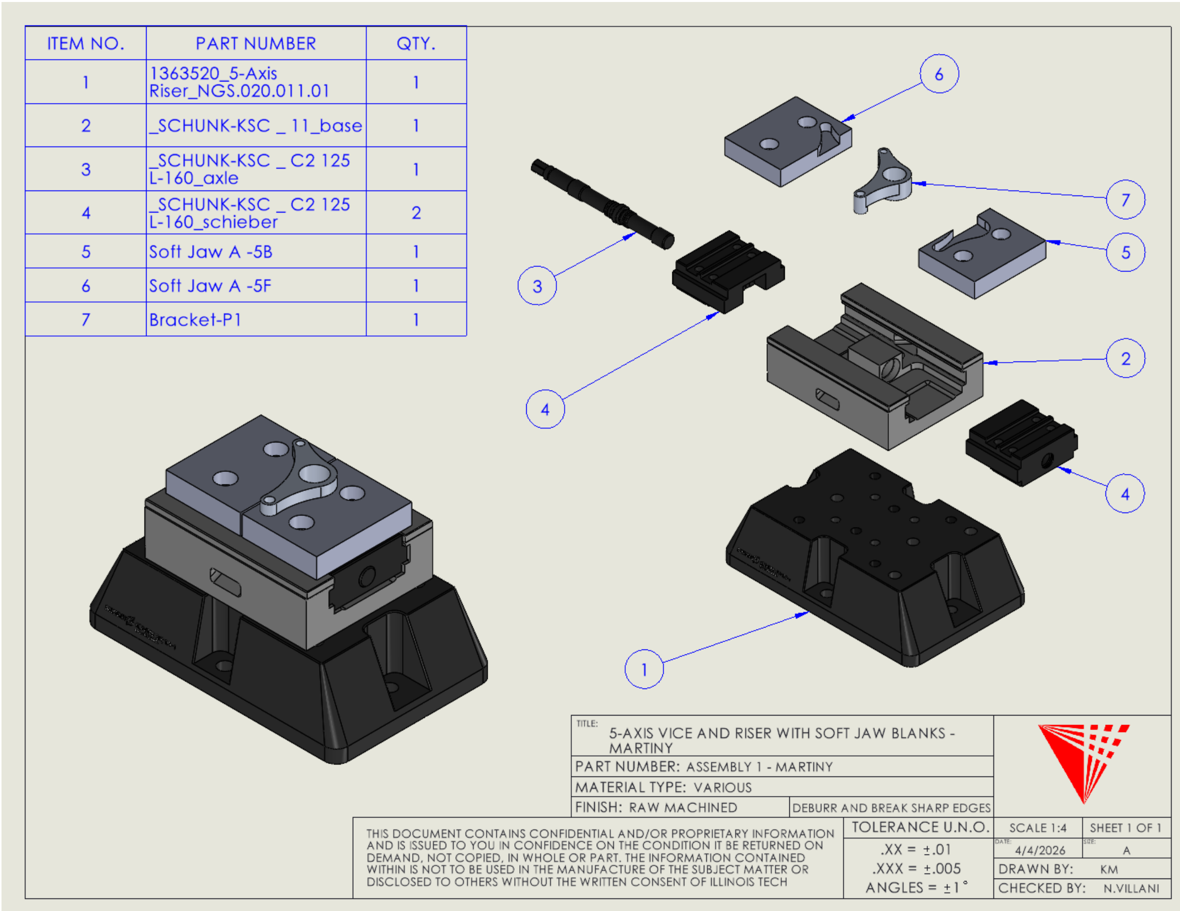



*Figure 34: Vice and riser exploded view and bill of materials*

# Mastercam P-1 Programming

The final part of this study was to program the manufacturing of both Bracket P-1 and Bracket P-2 using Mastercam with a supplied tool library. This limited the number of tools available for manufacturing and forced some decisions to be made which will be discussed alongside Bracket P-2.

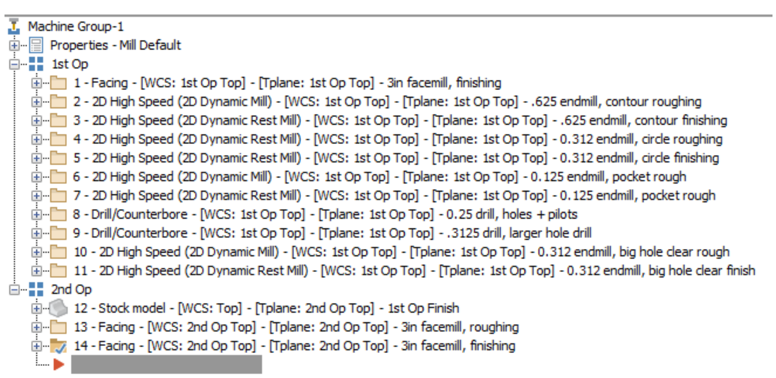



*Figure 35: Full toolpath list with descriptions for Bracket P-1*

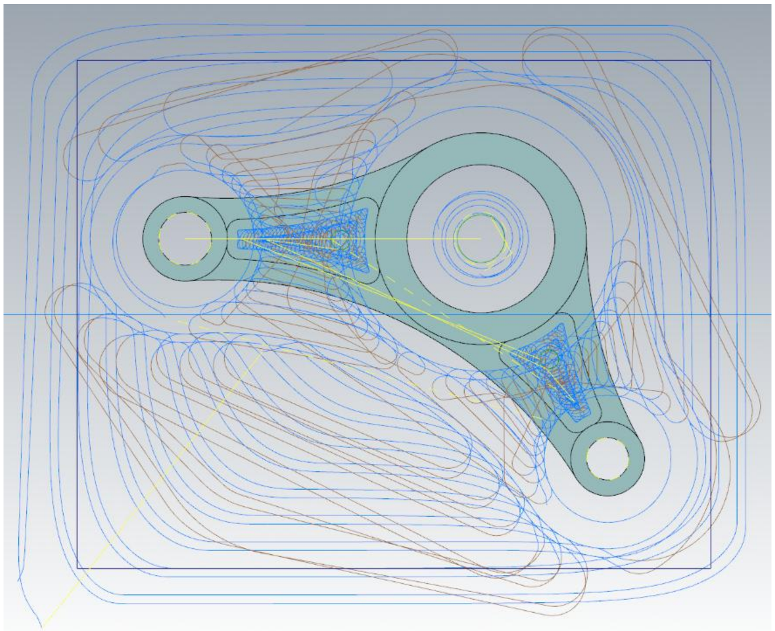


*Figure 36: Bracket P-1 first operation toolpaths*

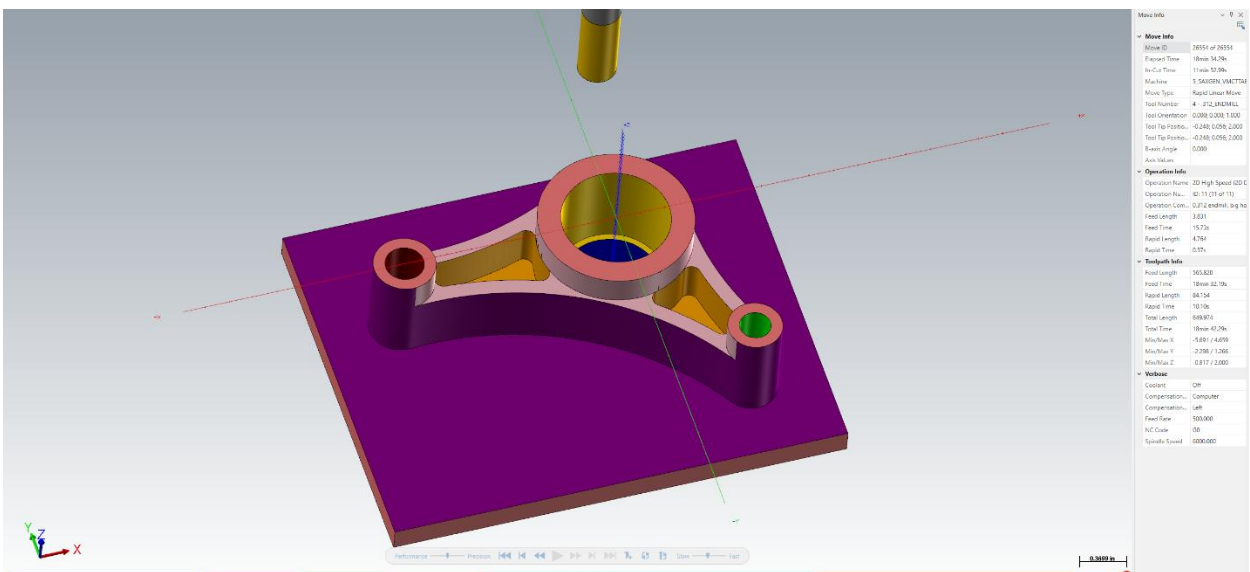



*Figure 37: Bracket P-1 end of first operation with color loop set to "By Operation"*

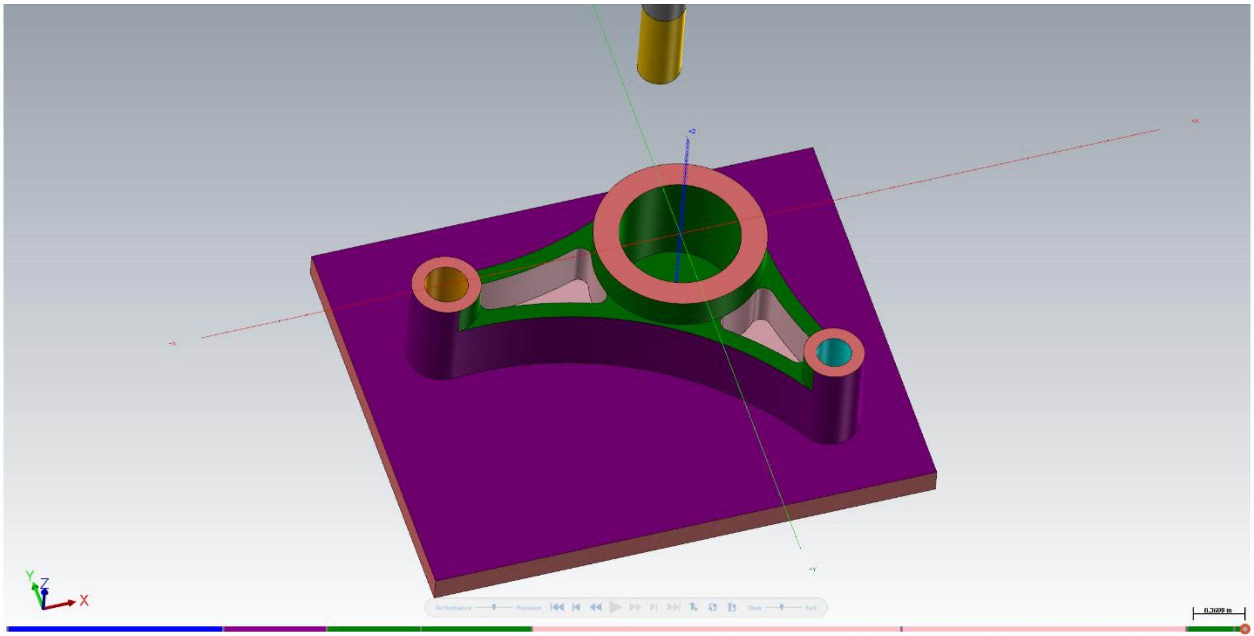


*Figure 38: Bracket P-2 end of first operation with color loop set to "By Tool"*

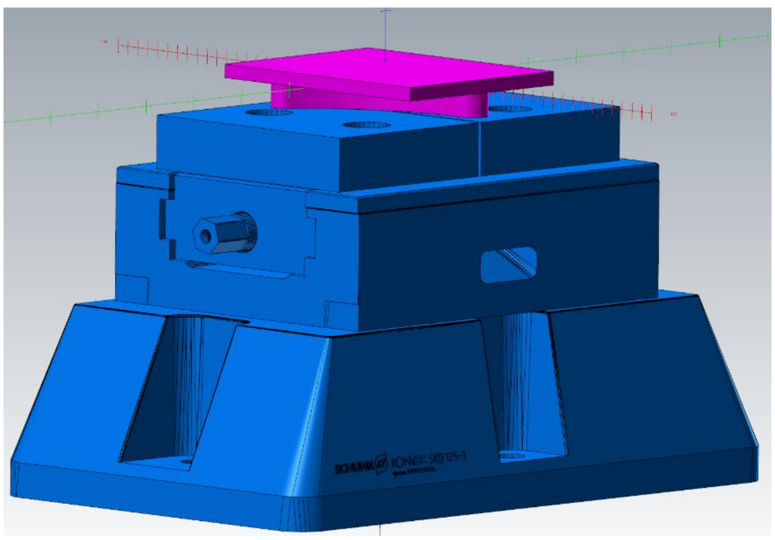


*Figure 39: Bracket P-1 set up for second operation with the vice and riser as a fixture and all of first operation used for stock material*

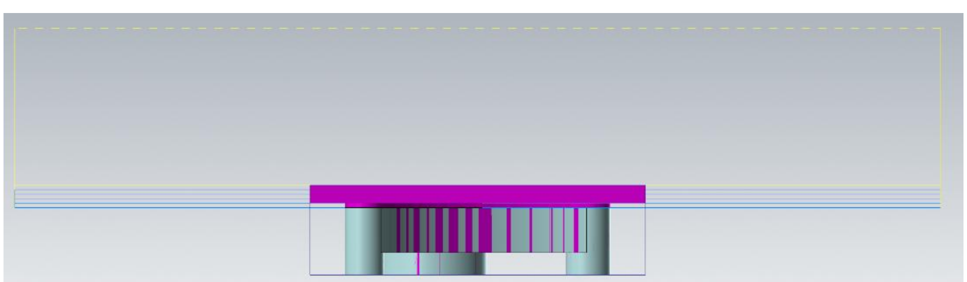


*Figure 40: Bracket P-1 second operation toolpaths*

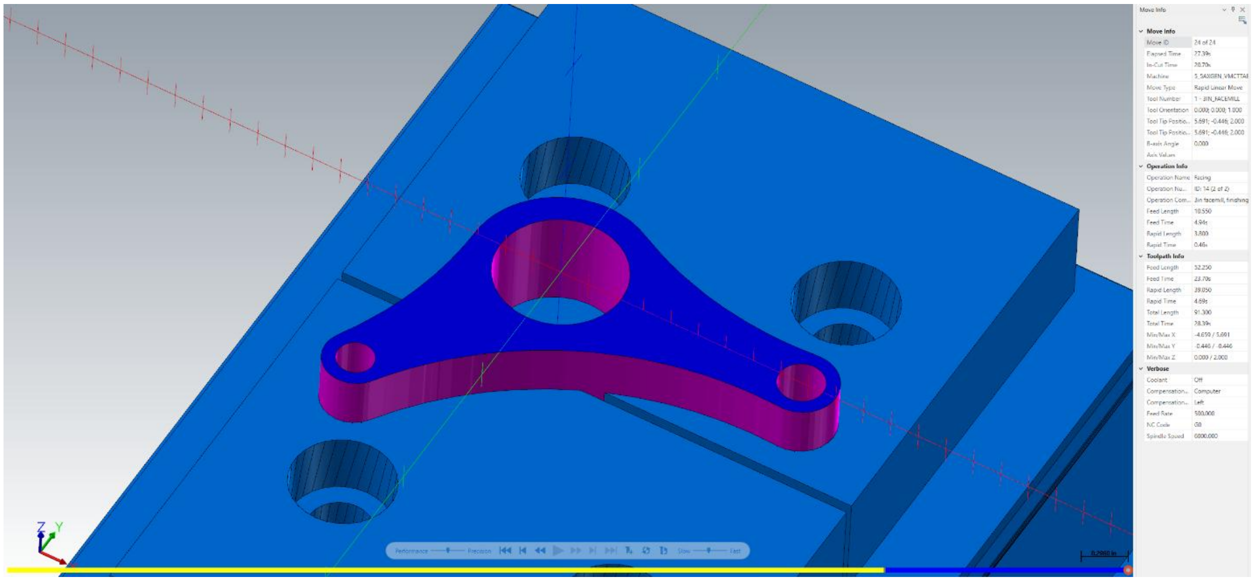


***Figure 41: Bracket P-1 end of second operation with color loop set to "By Operation"***

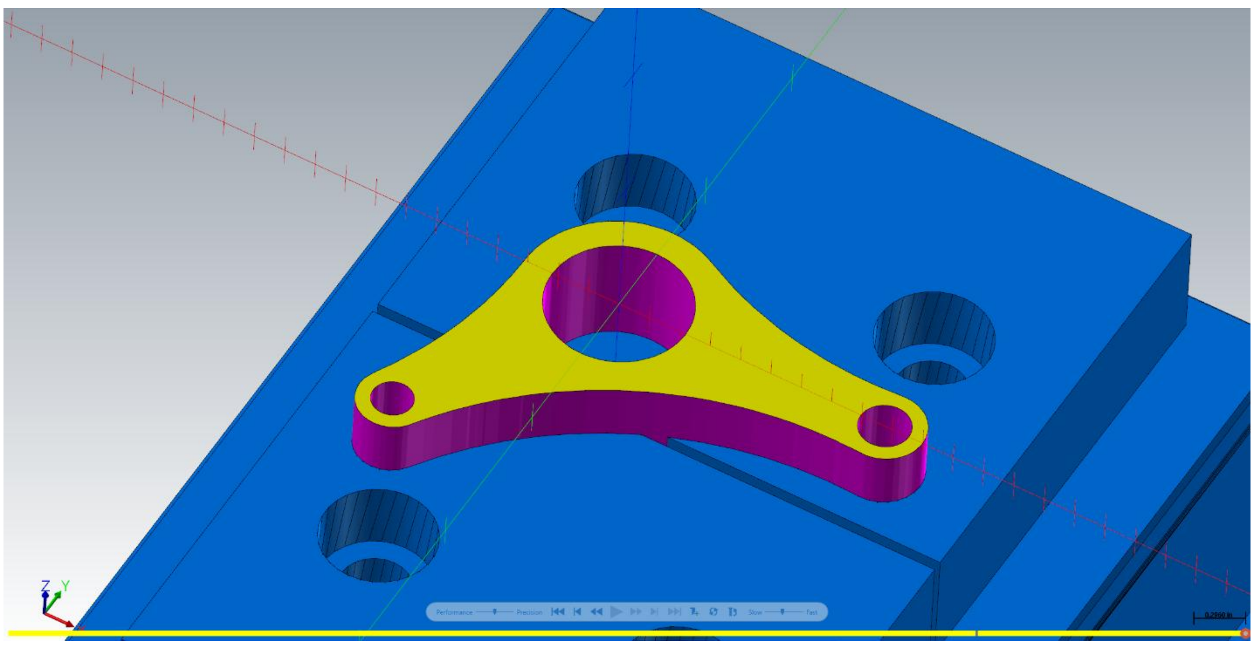


***Figure 42: Bracket P-1 end of second operation with color loop set to "By Tool"***

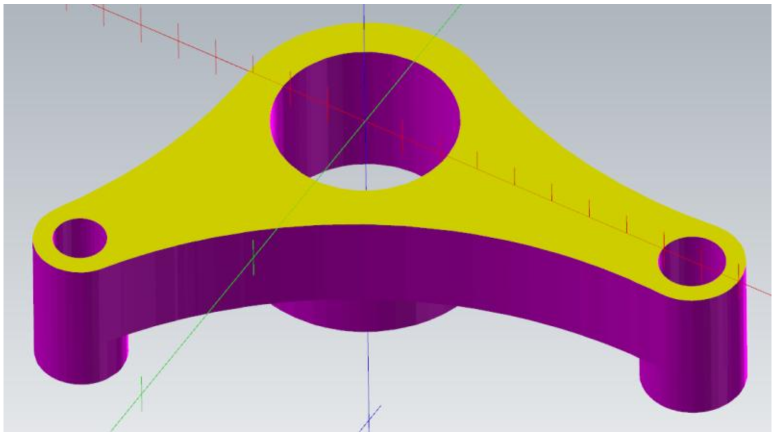


*Figure 43: Bracket P-1 top of second operation final part*

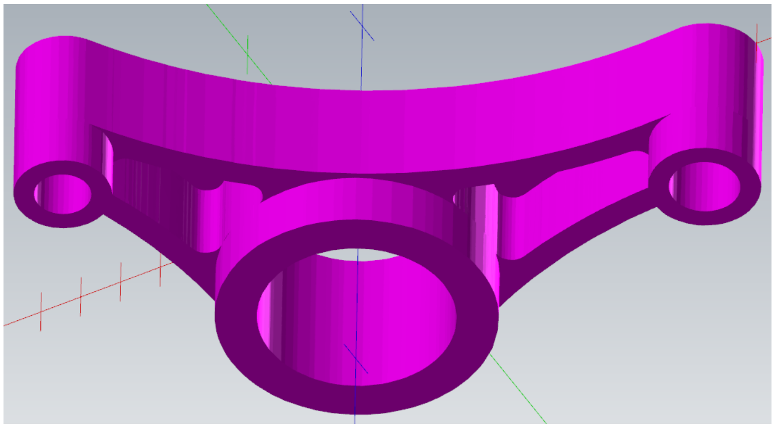


*Figure 44: Bracket P-1 bottom of second operation final part*

# Mastercam P-2 Programming


***Figure 45: Full toolpath list with descriptions for Bracket P-2***

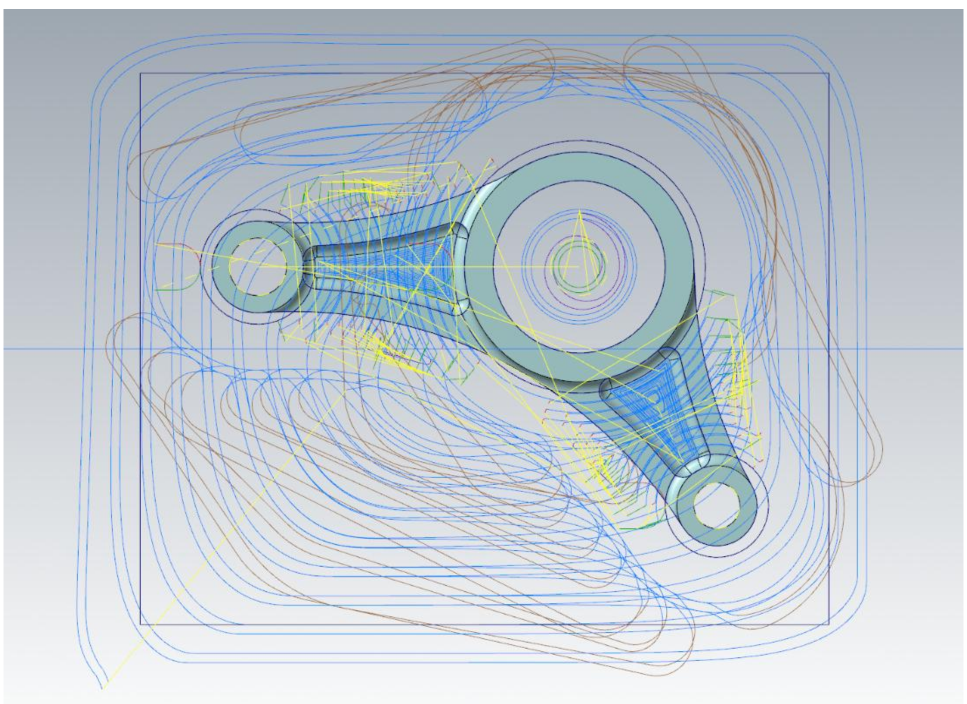


***Figure 46: Bracket P-2 first operation toolpaths***

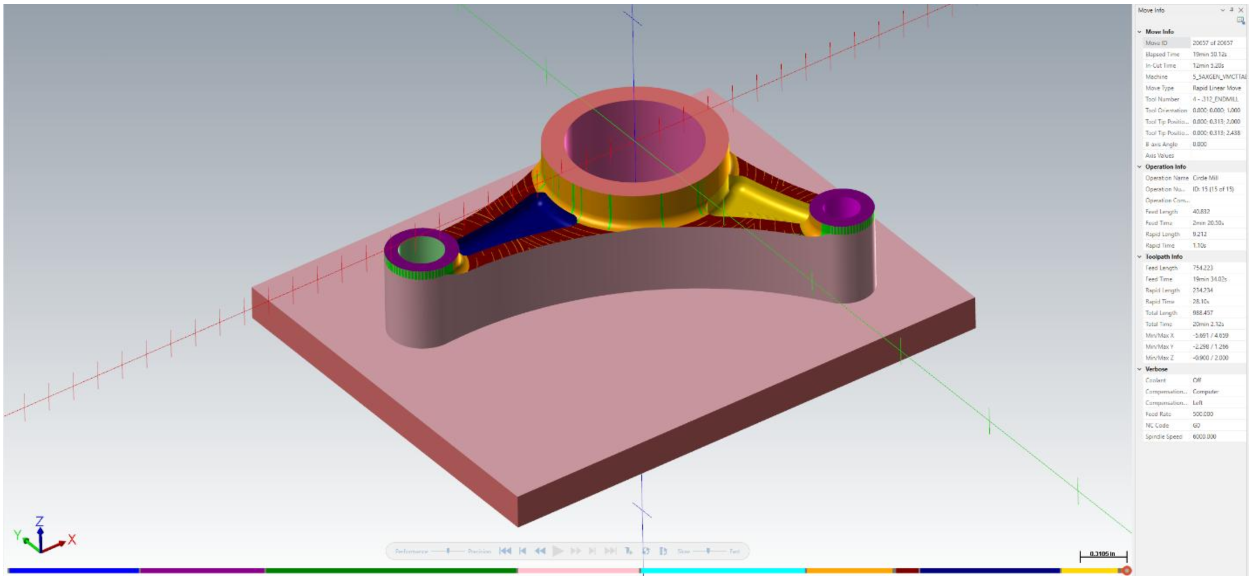


*Figure 47: Bracket P-2 end of first operation with color loop set to "By Operation"*

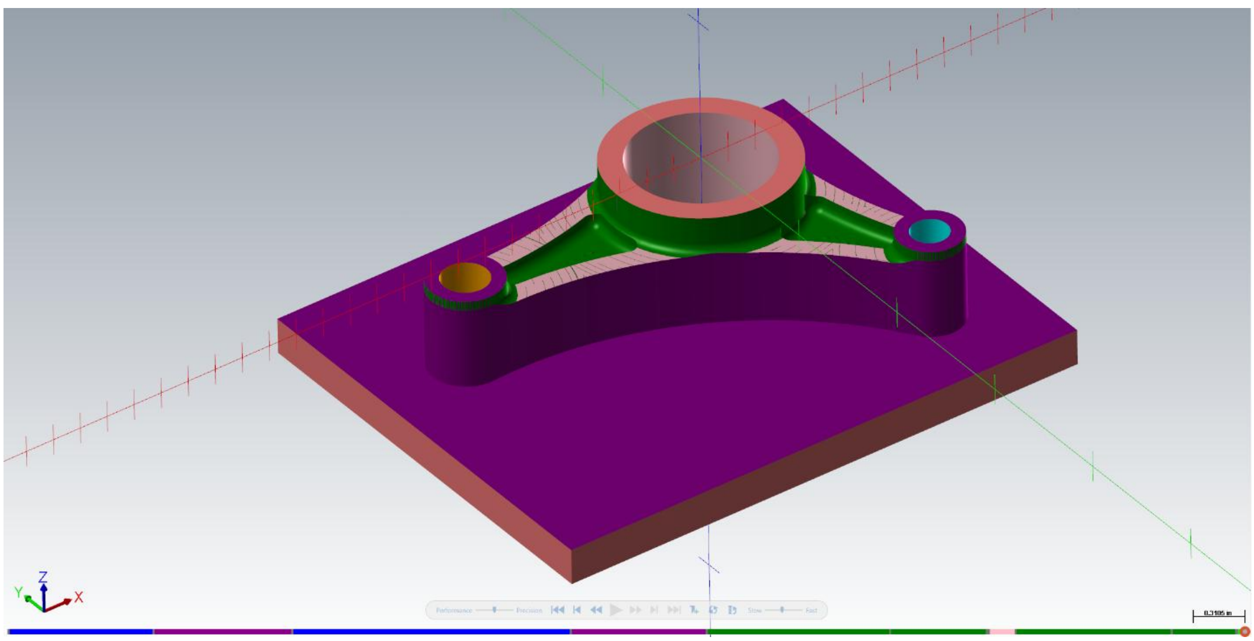


*Figure 48: Bracket P-2 end of first operation with color loop set to "By Tool"*

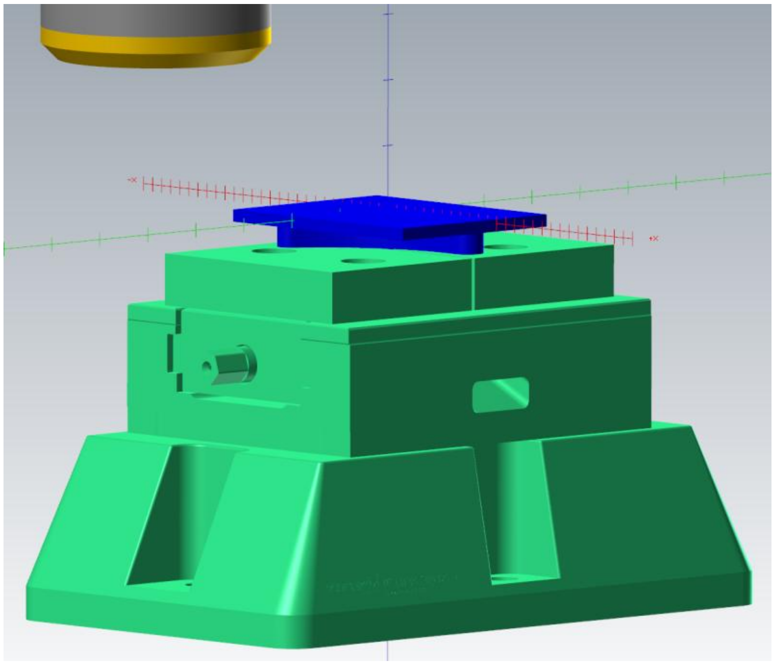


*Figure 49: Bracket P-2 set up for second operation with the vice and riser as a fixture and all of first operation used for stock material*

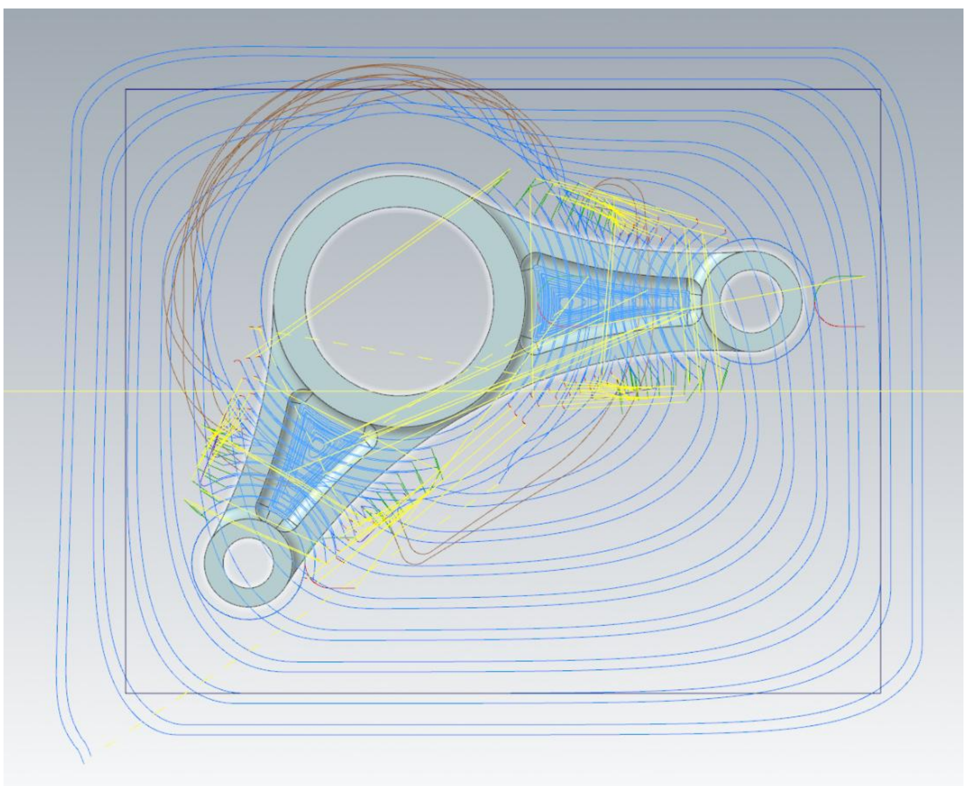


*Figure 50: Bracket P-2 second operation toolpaths*

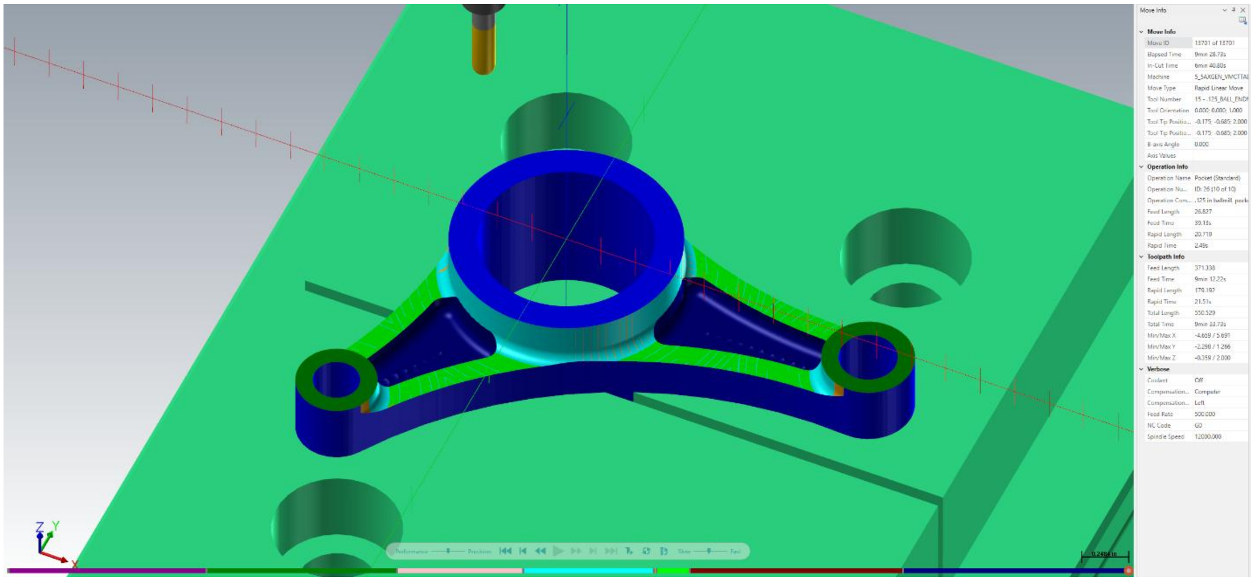


*Figure 51: Bracket P-2 end of second operation with color loop set to "By Operation"*

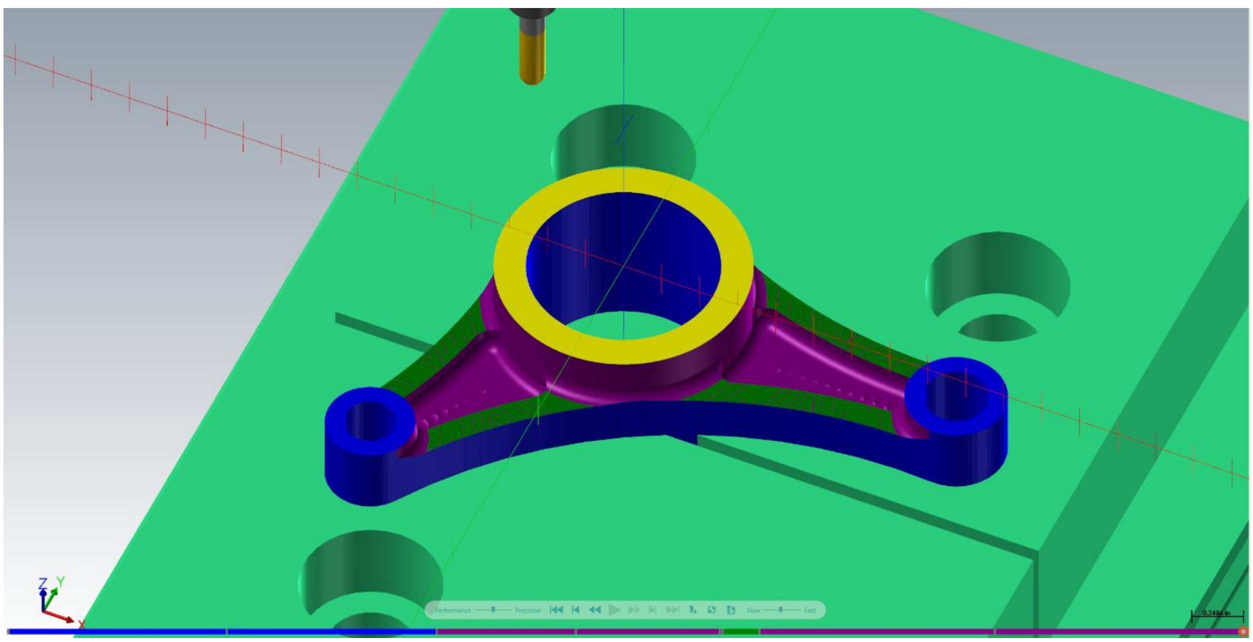


*Figure 52: Bracket P-2 end of second operation with color loop set to "By Tool"*

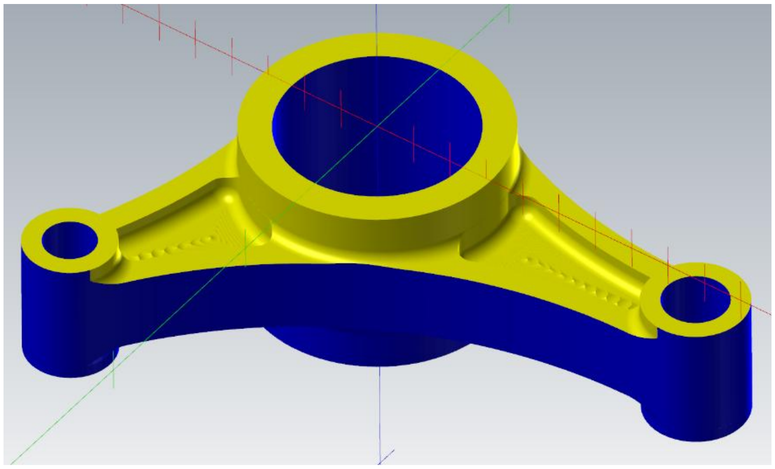


*Figure 53: Bracket P-2 top of second operation final part*

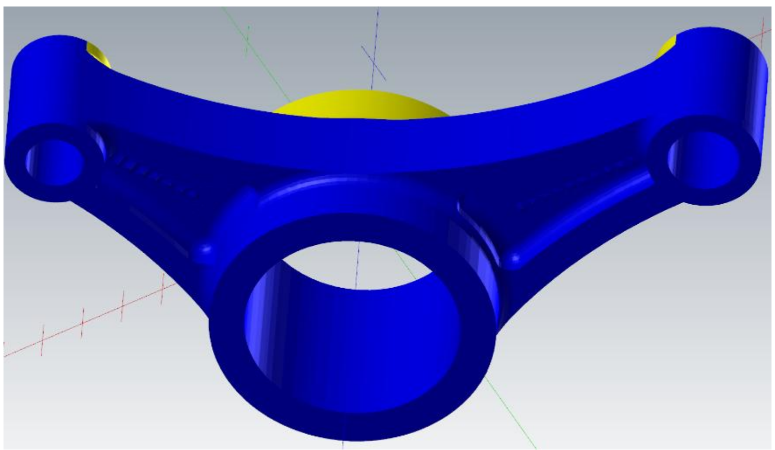


*Figure 54: Bracket P-2 bottom of second operation final part*

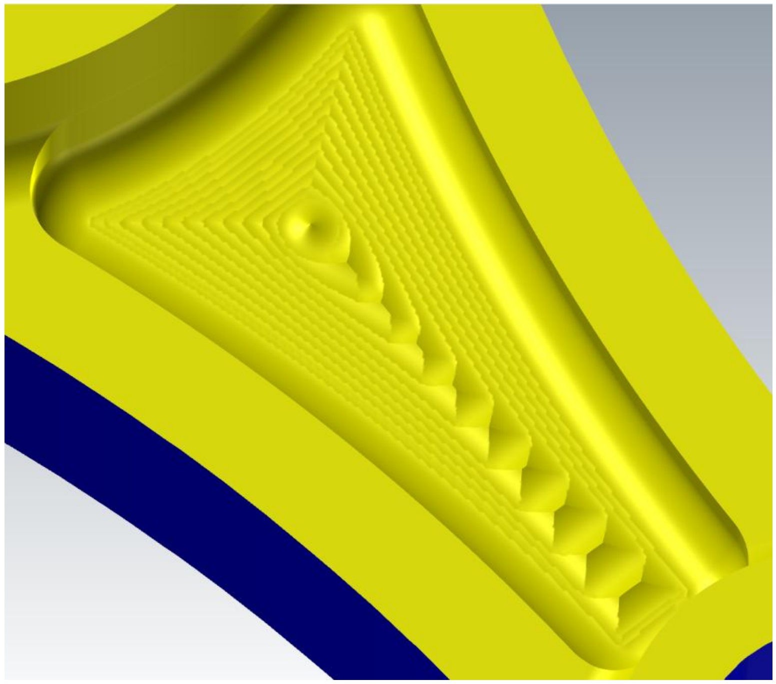


*Figure 55: Bracket P-2 pocket with ball mill*

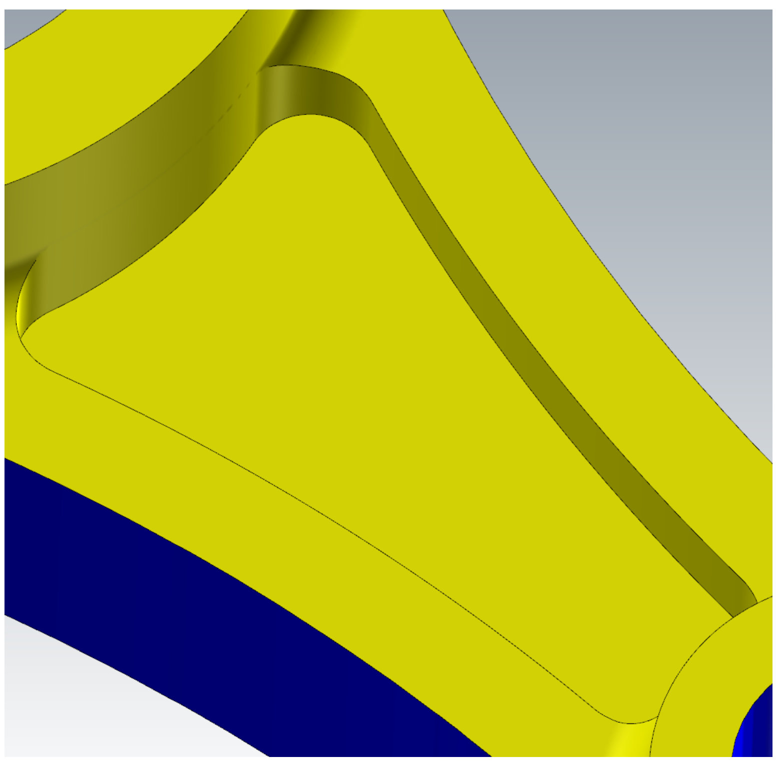


*Figure 56: Bracket P-2 pocket with endmill*

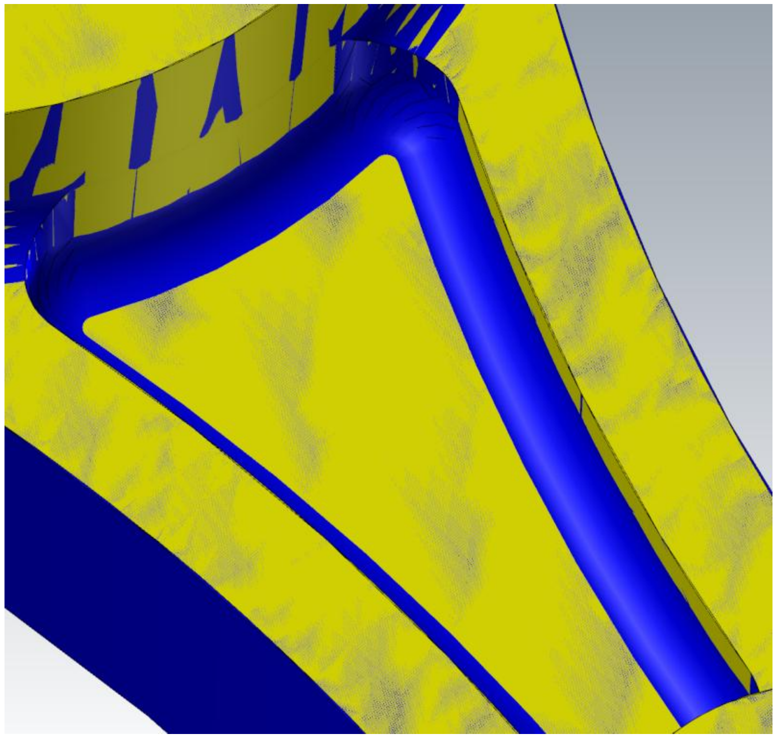


*Figure 57: Bracket P-2 pocket with bullnose endmill*

Due to the limited availability of tools for this case study, a manufacturing choice needed to be made in terms of the pockets. As seen in Figures 55, 56, and 57, the tool choice significantly altered the outcome of the pocket. In Figure 55 a ball mill was used. This resulted in a pocket that had the correct corner and floor radii, but a rough surface finish. This is due to the shape of a ball mill, which causes the ball to be tangent to a flat surface, making smooth finishes impossible to achieve. The surface finish issue can be fixed, as seen in Figure 56, with an endmill. An endmill maintains the corner radii and smooths the floor, but eliminates the floor radii, which would significantly alter the strength of the final part. In Figure 57, a bullnose endmill is used, which keeps the corner radii and floor smoothness, but decreases the floor radii to smaller than desired. For Figure 57, the final part in blue is overlayed on the milled-out material in yellow. In places where the milled and final part match, the overlap between the two is clear, such as in the top left corner where the large circle is. This overlap is not present in the pocket floor corners. Because blue is dominant along the floor, that means that more material was milled from the stock than should have been. For the floor radius to be milled with a bullnose endmill, a larger tool is needed. While this would fix the floor radius, the increase in diameter of the tool would prevent the corner radii from being milled. From each of these options, a 0.125-inch ball mill was used for the final part.

When using a ball mill, the smallest floor radii that can be achieved is half of the tool's diameter. Using a 0.125-inch tool, which gives a floor radius of 0.0625 inches. For the corner radii, they can be machined if the tool is smaller than the radius of the corner. If the tool is the same size as the radius, then the tool will be fully engaged and can lead to tool chatter.

Bracket P-1 was easier to manufacture, taking less toolpaths at 14 total compared to Bracket P-2's 26 toolpaths. Bracket P-1 also did not have the issue with the pocket and was able to be milled with an endmill. Bracket P-1 took 19 minutes and 1 second of total mill time with 12 minutes and 12 seconds spent cutting. Bracket P-2 took 29 minutes and 18 seconds with 18 minutes and 45 seconds spent cutting. While Bracket P-2 took longer to manufacture, it was also a significantly stronger part